

The Effect of a Mediterranean Diet on the Cognitive and Physiological Symptoms of Premenstrual Dysphoric Disorder

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INTRODUCTION

Premenstrual dysphoric disorder (PMDD) is a severe form of premenstrual syndrome (PMS) that affects 3-8% of menstruating women. [1] It is a cyclical condition characterised by debilitating mood and physical symptoms that significantly impair daily functioning during the luteal phase of the menstrual cycle. [1]

Whilst its exact aetiology remains unclear, proposed mechanisms include altered sensitivity to fluctuations in the progesterone metabolite allopregnanolone and GABA-A receptor signalling, systemic inflammation, genetic and epigenetic predisposition, and dysregulation of the gut-brain axis. [2,3] The Mediterranean diet is rich in omega-3 fatty acids, fibre, and antioxidants, which may influence several of these pathways, particularly inflammation and gut-brain signalling. [4]

Current management primarily involves pharmacological and hormonal therapies, including continuous dosing of combined oral contraceptives, selective serotonin reuptake inhibitor (SSRI) dosing prior to menses, and in severe cases, gonadotrophin-releasing hormone (GnRH) analogues to cease ovulation. [2] However, there is limited evidence regarding dietary approaches for managing PMDD symptoms.

This study aims to determine whether a 12-week Mediterranean diet intervention reduces PMDD symptom severity compared with a control condition, and whether any symptom improvement is associated with changes in systemic inflammation, measured using high-sensitivity C-reactive protein (hs-CRP).

METHODS

A 12-week longitudinal, parallel-group randomised controlled trial will be conducted in 50 Australian women aged 18-40 with untreated PMDD. Participants will be randomly allocated to either a Mediterranean diet intervention (n=25) or control group (n=25). Participants will commence the study within the first three days of their menstrual cycle. At baseline, all participants will undergo blood collection for hs-CRP. The intervention group will also receive an initial dietitian consultation to support adoption of a Mediterranean diet, followed by consultations at weeks

2, 4 and 8. Mediterranean diet adherence will be assessed in both groups using the 14-item Mediterranean Diet Adherence Screener (MEDAS) at baseline and week 12, with additional assessments conducted in the intervention group at weeks 2, 4, 6 and 8. Control participants will maintain their usual diet and will not receive dietary counselling.

Throughout the 12-week study, both groups will complete the Daily Record of Severity of Problems each evening, rating PMDD symptoms from 1 (not at all) to 6 (extreme). At week 12, participants will complete a three-day food diary and repeat hs-CRP testing. Daily PMDD symptom severity and hs-CRP concentrations will be compared within and between groups across the 12-week intervention to determine the effect of the Mediterranean diet on symptoms and systemic inflammation.

EXPECTED OUTCOME & SIGNIFICANCE

It is anticipated that participants in the Mediterranean diet group will demonstrate improvements in PMDD symptom severity and reductions in hs-CRP by the end of the 12-week intervention compared with baseline, with greater improvements than those observed in the control group. These findings would provide preliminary evidence that a Mediterranean diet may be an effective non-pharmacological strategy for reducing PMDD symptoms and may help clarify the potential role of inflammation in symptom severity. This could support further investigation of dietary interventions as an adjunct to existing PMDD treatments

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